

```

1  #include <stdio.h>
2  int main() {
3      int nof, nor;
4      printf("Enter number of frames: ");
5      scanf("%d", &nof);
6      printf("Enter number of pages: ");
7      scanf("%d", &nor);
8      int pages[50];
9
10     printf("Enter reference string:\n");
11
12     for (int i = 0; i < nor; i++) scanf("%d", &pages[i]);
13
14     int frame[10];
15
16     for (int i = 0; i < nof; i++) frame[i] = -1;
17
18     int pageFaults = 0;
19
20     printf("\nPage\tFrames\n");
21
22     for (int i = 0; i < nor; i++) {
23         int found = 0;
24
25         // Check page hit
26         for (int j = 0; j < nof; j++) {
27             if (frame[j] == pages[i]) {
28                 found = 1;
29                 break;
30             }
31         }
32
33         if (!found) {
34             int pos = -1;
35
36             // Find empty frame
37             for (int j = 0; j < nof; j++) {
38                 if (frame[j] == -1) {
39                     pos = j;
40                     break;
41                 }
42             }
43
44             // Need replacement
45             if (pos == -1) {
46                 int farthest = -1;
47
48                 for (int j = 0; j < nof; j++) {
49                     int k;
50
51                     for (k = i + 1; k < nor; k++) {
52                         if (frame[j] == pages[k]) break;
53                     }
54
55                     // Never used again
56                     if (k == nor) {
57                         pos = j;
58                         break;
59                     }
60
61                     if (k > farthest) {
62                         farthest = k;
63                         pos = j;
64                     }
65                 }
66             }
67
68             frame[pos] = pages[i];
69             pageFaults++;
70         }
71
72         printf("%d\t", pages[i]);
73         for (int j = 0; j < nof; j++) printf("%d ", frame[j]);
74         printf("\n");
75     }
76     printf("\nTotal Page Faults = %d\n", pageFaults);
77 }

```